

Aashish Awadhani

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SUMMARY

Machine Learning Engineer with expertise in developing real-time computer vision systems and Large Language Model (LLM) applications. My proven track record includes engineering OCR pipelines and high-performance object detection stacks on Jetson AGX Orin, resulting in a 40% reduction in latency. I am experienced in end-to-end AI pipelines, encompassing model optimization (quantization, pruning), agentic workflows, and deploying scalable RAG-based solutions.

EXPERIENCE

CalSys Lab, Cal Poly Pomona | *AI/ML Research Assistant*

Sep 2025 - Present

- Built and benchmarked 4 OCR pipelines (CNN, YOLO-based detection, CRNN-CTC, and Transformer OCR) for CAPTCHA recognition, significantly improving sequence accuracy across noisy and distorted inputs.
- Developed character recognition models using PyTorch and TensorFlow, leveraging CNNs and Vision Transformers (ViTs) to maximize recognition accuracy on distorted character sets.
- Optimized character segmentation by implementing a hybrid YOLO + CRNN architecture, reducing segmentation errors across complex datasets.

Autonomous Vehicle Lab, Cal Poly Pomona | *Machine Learning Engineer*

May 2024 - May 2025

- Integrated YOLOv8 and YOLOv11 with ROS2 to enhance multi-sensor perception, fusing RGB, LiDAR, and depth data for robust environmental mapping.
- Executed model compression techniques, including quantization, pruning, and ONNX/TensorRT optimization, enabling reliable deployment under strict embedded hardware constraints.
- Leveraged MLflow to track experiments and tune hyperparameters based on real-world sensor data, reducing model retraining cycles and improving detection precision.

EDUCATION

California State Polytechnic University | *Master of Science, Computer Science* (GPA: 3.97)

Pomona Aug 2023 - May 2025

PROJECTS

GestureLink (Real-Time Hand Pose Analysis)

- Architected a decoupled macOS/iOS system using the Apple Vision framework to perform real-time hand pose analysis and gesture recognition.
- Optimized inference speeds for mobile deployment, ensuring responsive gesture tracking with minimal computational overhead and battery consumption.

Agentic Marketing (Multi-Agent AI Workflow)

- Engineered an automated marketing workflow using CrewAI to orchestrate four collaborative agents for strategy generation, SEO-copywriting, and content calendar management.
- Implemented asynchronous task execution powered by Gemini 1.5/2.0 Flash to handle high-volume content generation while maintaining high reasoning quality and cost-efficiency.

Real-Time Chat Application

- Built a responsive full-stack platform using the MERN stack and Socket.io for live bi-directional messaging, featuring secure JWT-based authentication.
- Managed global application state via Zustand and implemented a modern, mobile-responsive UI using Tailwind CSS.

CERTIFICATION

- Rapid Application Development with LLMs NVIDIA: Developed production-grade LLM applications using Transformers, LangChain, and Vector Databases for RAG and semantic search pipelines.

SKILLS

- Machine Learning & AI:** PyTorch, TensorFlow, TensorRT, YOLO, OpenCV, CNN, RNN, Vision Transformers, Transformers, ONNX, Quantization, Pruning, CUDA, MLflow, Large Language Models, Hugging Face, LangChain, RAG, Vector DB, Natural Language Processing, Generative AI, Diffusion Models, AI Agents, OpenAI API, LLMops, APIs
- Software Engineering & Systems:** Python, C++, Java, JavaScript, Node.js, Express.js, Rest APIs, Docker, Git, Linux, ROS2, Fast API, MySQL, PostgreSQL, MongoDB, Redis, Swift
- Tools & Cloud:** AWS, GCP, Azure, GitHub Actions, PowerBI, Tableau, Streamlit, PyPDF2, CI/CD